

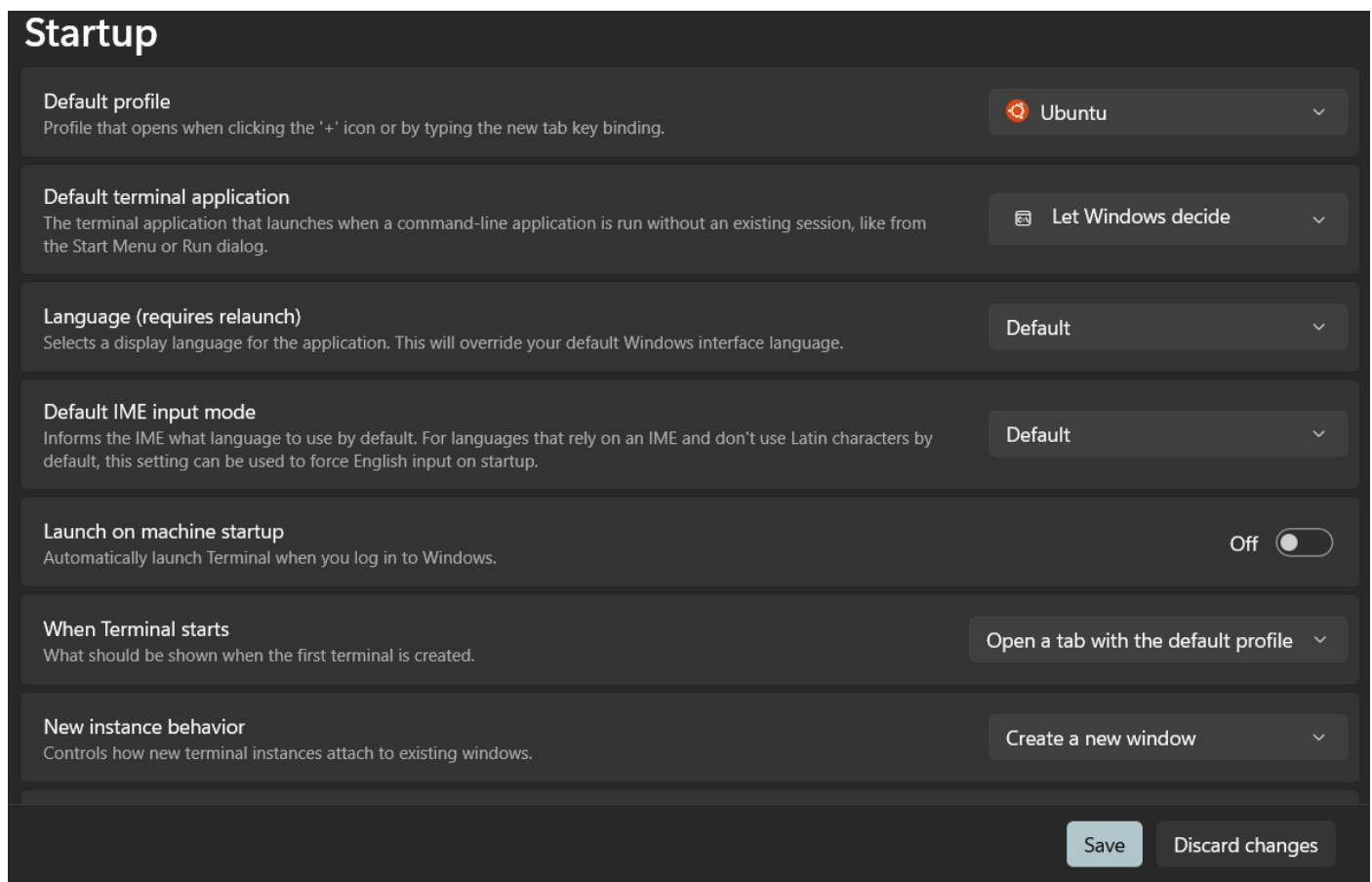
How to install Linux on Windows with WSL

1. Open PowerShell in administrator mode by right-clicking and selecting "**Run as administrator**", and enter this below command.

```
wsl --install
```

2. then **restart your machine**.

Now search for ubuntu app and open it. press ctrl + , and then set the **Default profile** as **Ubuntu** as shown in the image and save.



Installing Docker

Firstly follow these steps:

1. Set up Docker's apt repository by running these commands (one-by-one) in a new terminal:

Add Docker's official GPG key:

```
sudo apt-get update
sudo apt-get install ca-certificates curl
sudo install -m 0755 -d /etc/apt/keyrings
sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
sudo chmod a+r /etc/apt/keyrings/docker.asc
```

Add the repository to Apt sources: (fully copy the below box and paste at a same time)

```
echo \  
"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc]  
https://download.docker.com/linux/ubuntu \  
$(. /etc/os-release && echo "$VERSION_CODENAME") stable" | \  
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

```
sudo apt-get update
```

2. Install the Docker packages by running the following command in the same terminal:

```
sudo apt-get install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
```

3. Verify your installation by running this command:

```
sudo docker run hello-world
```

Follow these next steps to be able to run docker commands without "sudo"

1. Create a docker user group.

```
sudo groupadd docker
```

2. Add your user to the docker group.

```
sudo usermod -aG docker $USER
```

3. Log out and log back in to your system, or run the following command:

```
newgrp docker
```

4. Test that you can run docker commands without sudo.

```
docker run hello-world
```

(Optional) If you are running Docker on a Linux machine equipped with an Nvidia GPU and the proper installed drivers

Install the Nvidia Container Toolkit by following these steps:

1. Configure the production repository: (copy the below box at the same time

```
curl -fsSL https://nvidia.github.io/libnvidia-container/gpgkey | sudo gpg --dearmor -o /usr/share/keyrings/nvidia-container-toolkit-keyring.gpg \
&& curl -s -L https://nvidia.github.io/libnvidia-container/stable/deb/nvidia-container-toolkit.list | \
sed 's#deb https://#deb [signed-by=/usr/share/keyrings/nvidia-container-toolkit-keyring.gpg] https://#g' | \
sudo tee /etc/apt/sources.list.d/nvidia-container-toolkit.list
```

2. Optionally, configure the repository to use experimental packages:

```
sed -i -e '/experimental/ s/^#//g' /etc/apt/sources.list.d/nvidia-container-toolkit.list
```

2. Update the packages list from the repository:

```
sudo apt-get update
```

3. Install the NVIDIA Container Toolkit packages:

```
sudo apt-get install -y nvidia-container-toolkit
```

Configure the Nvidia Container Toolkit for Docker:

1. Configure the container runtime by using the nvidia-ctk command:

```
sudo nvidia-ctk runtime configure --runtime=docker
```

2. Restart the Docker daemon:

```
sudo systemctl restart docker
```

Installation

1. Clone the repository:

```
git clone https://github.com/henki-robotics/robotics_essentials_ros2.git
```

2. Create a new workspace for your exercises. This will be automatically mounted and available from the Docker container

```
mkdir -p $HOME/exercises_ws/src
```

Launching the Docker container

Before anything run the following command in a new terminal. This will give display permissions to your containers:

```
sudo apt install x11-xserver-utils  
  
xhost +
```

Note: This has to be run everytime you restart your computer

1. Use docker compose to build and run the Docker container, which includes the ROS 2, Gazebo and simulated Andino installation.

```
cd robotics_essentials_ros2/docker/  
  
docker compose up
```

Wait till it comes like "Attaching to robotics_essentials_ros2".

```
=> CACHED [robotics_essentials_ros2 16/18] WORKDIR /home/user/ros2_ws 0.0s  
=> CACHED [robotics_essentials_ros2 17/18] RUN sudo echo ". /opt/ros/humble/setup.bash" >> ~/.bashrc 0.0s  
=> CACHED [robotics_essentials_ros2 18/18] COPY ./ros_entrypoint.sh /ros_entrypoint.sh 0.0s  
=> [robotics_essentials_ros2] exporting to image 0.0s  
=> => exporting layers 0.0s  
=> => writing image sha256:4b3e6b741446e9693b69536951214b550f9e51aae124d93c306d8be2ac609db5 0.0s  
=> => naming to docker.io/library/robotics_essentials_ros2 0.0s  
=> [robotics_essentials_ros2] resolving provenance for metadata file 0.0s  
[+] Running 1/0  
✔ Container robotics_essentials_ros2 Created 0.0s  
Attaching to robotics_essentials_ros2
```

2. Don't close this above terminal where the docker container is running.
3. Open a new terminal with ALT and + and run this command to start a new terminal inside the Docker container:

```
docker exec -it robotics_essentials_ros2 bash
```